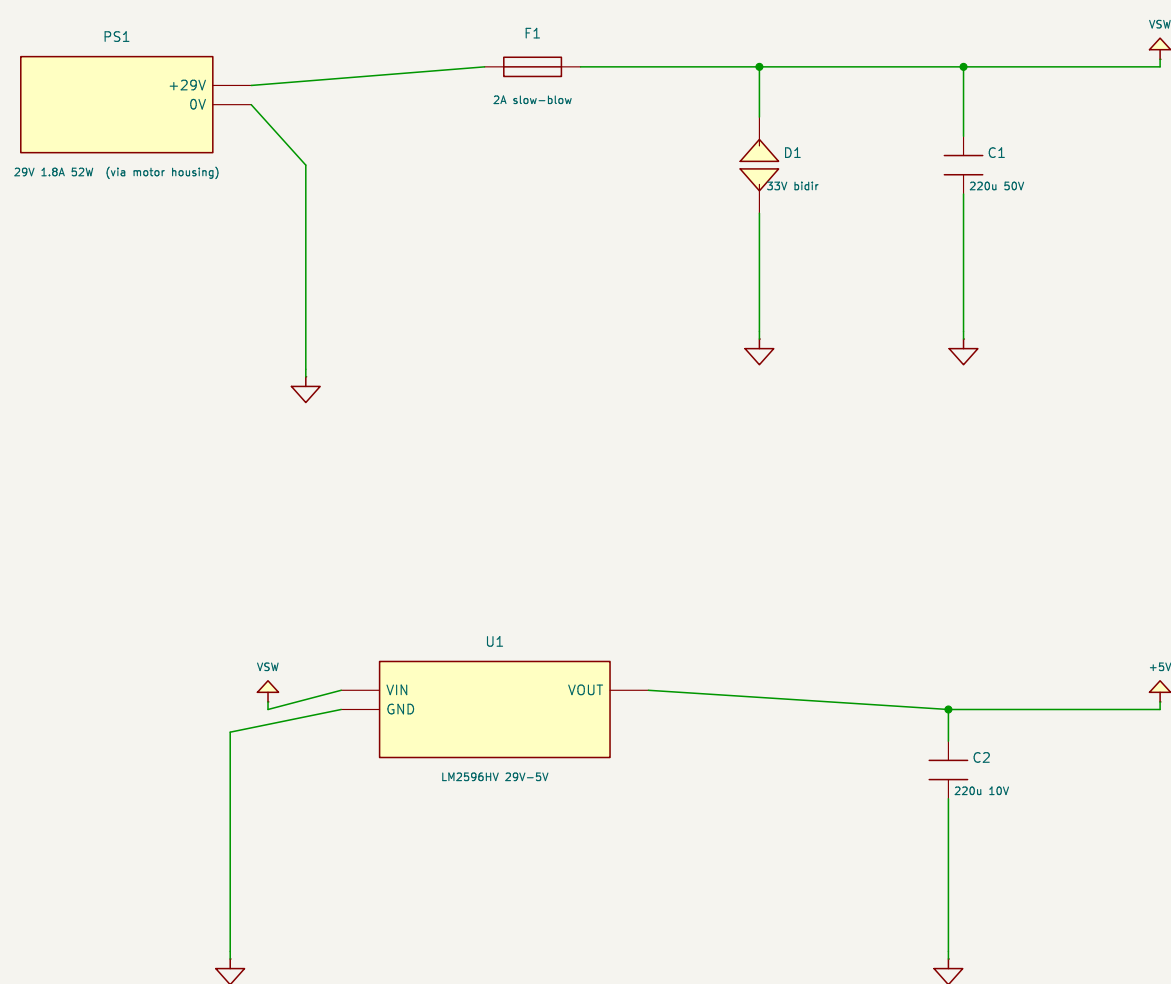
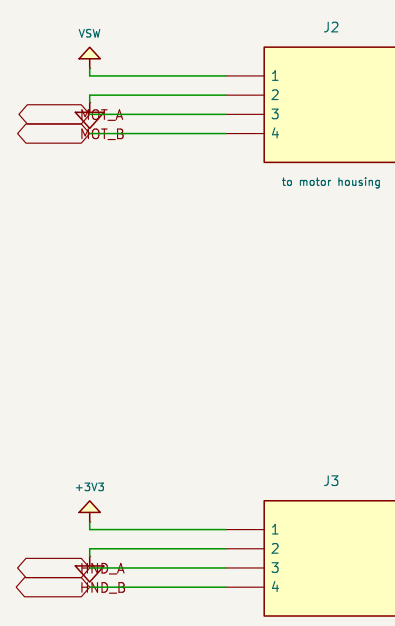


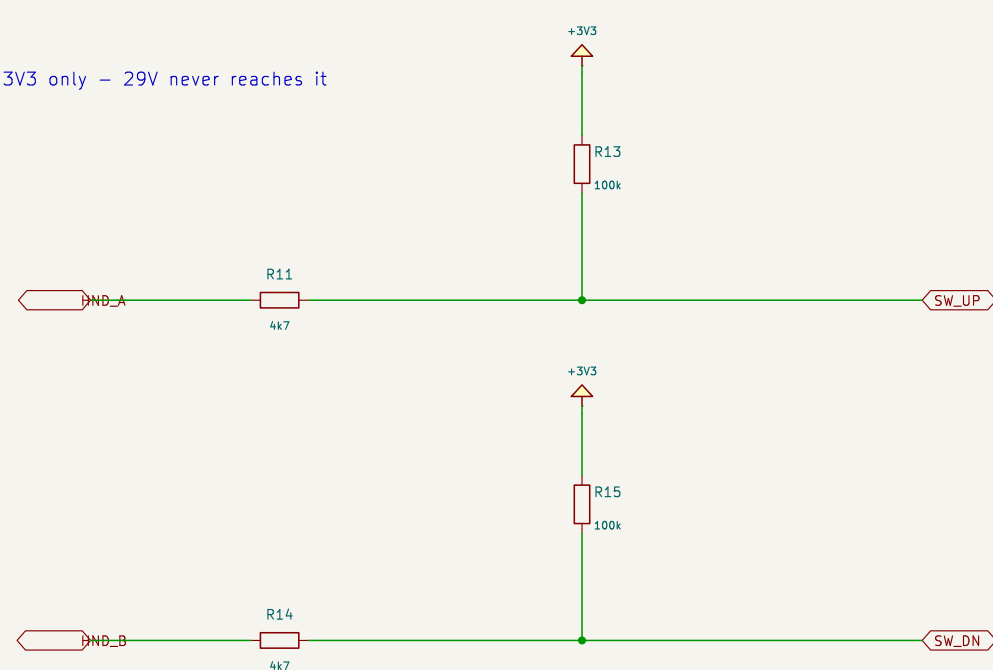
SUPPLY



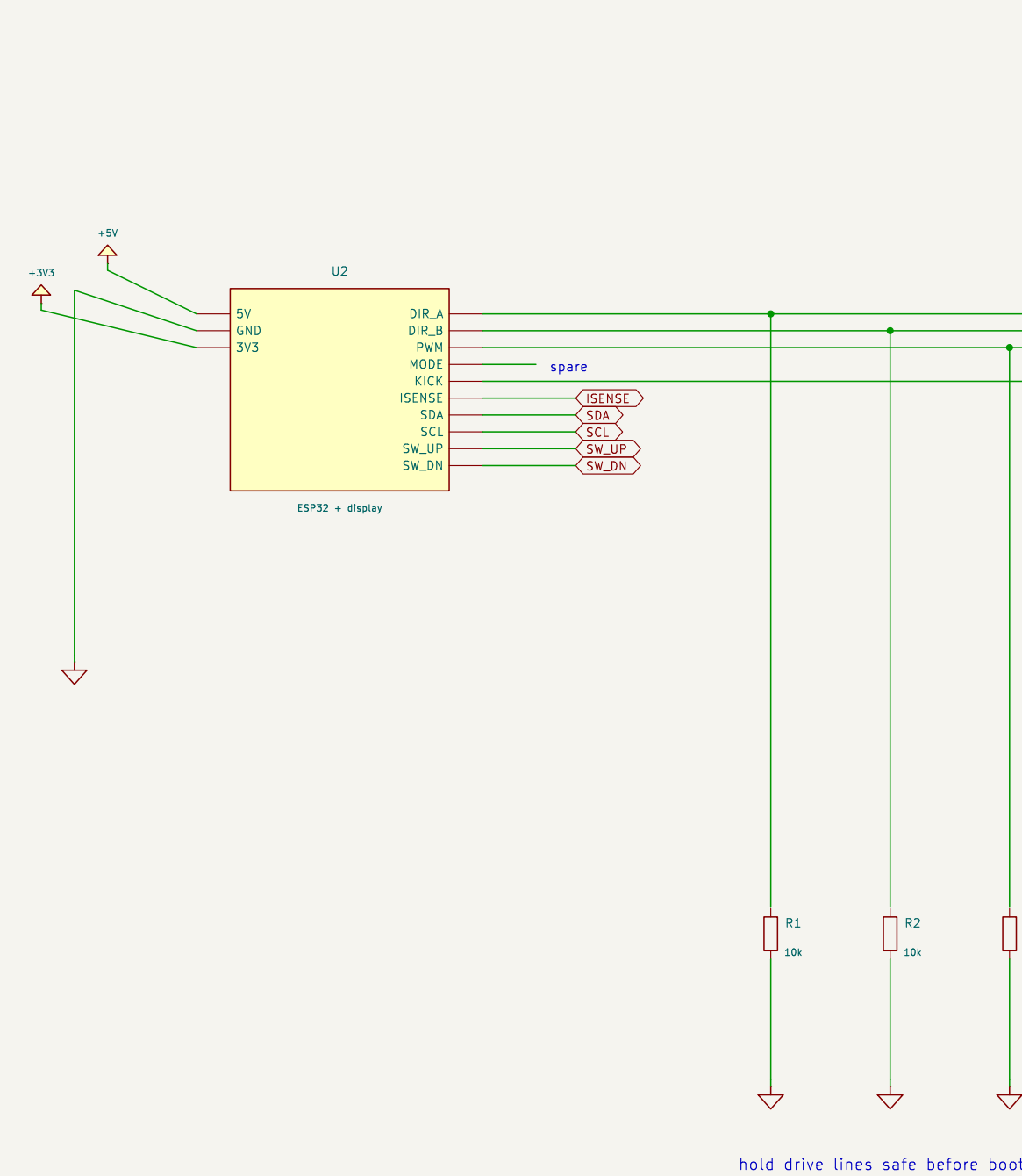
CABLE



handset sees 3V3 only – 29V never reaches it

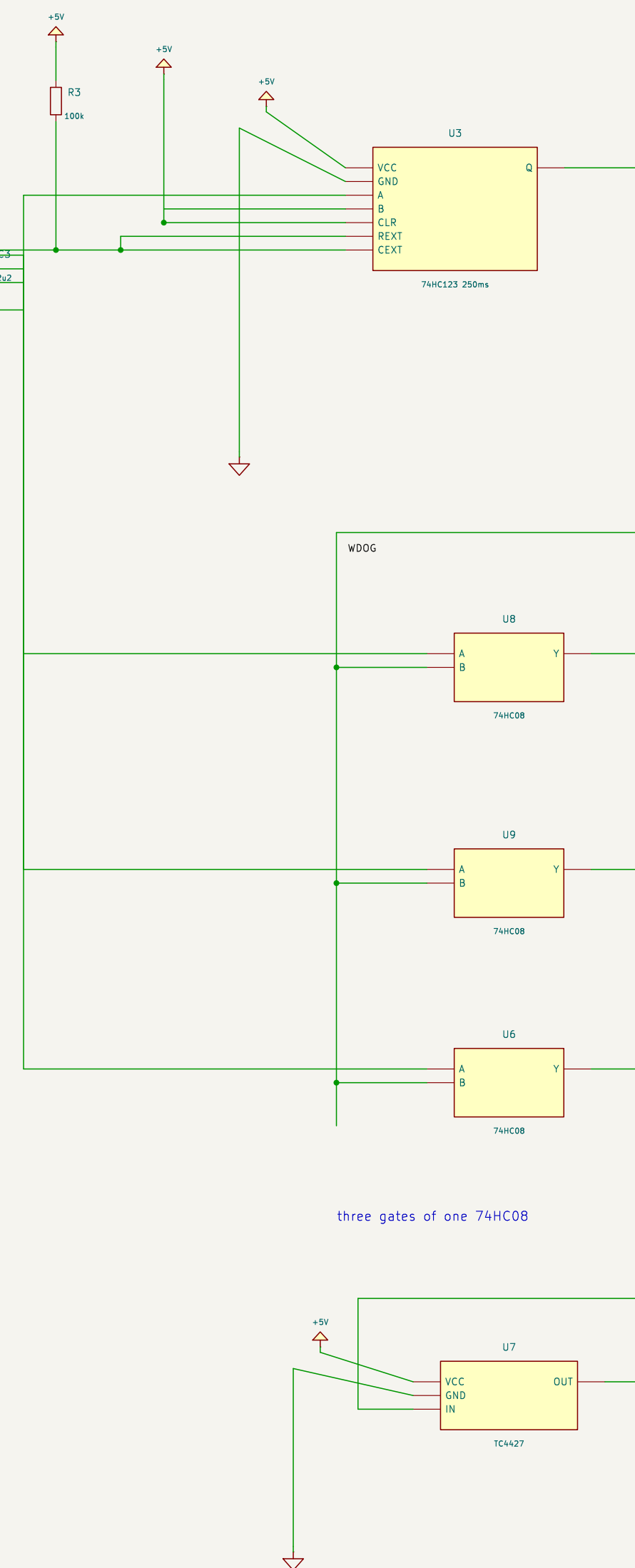


CONTROLLER



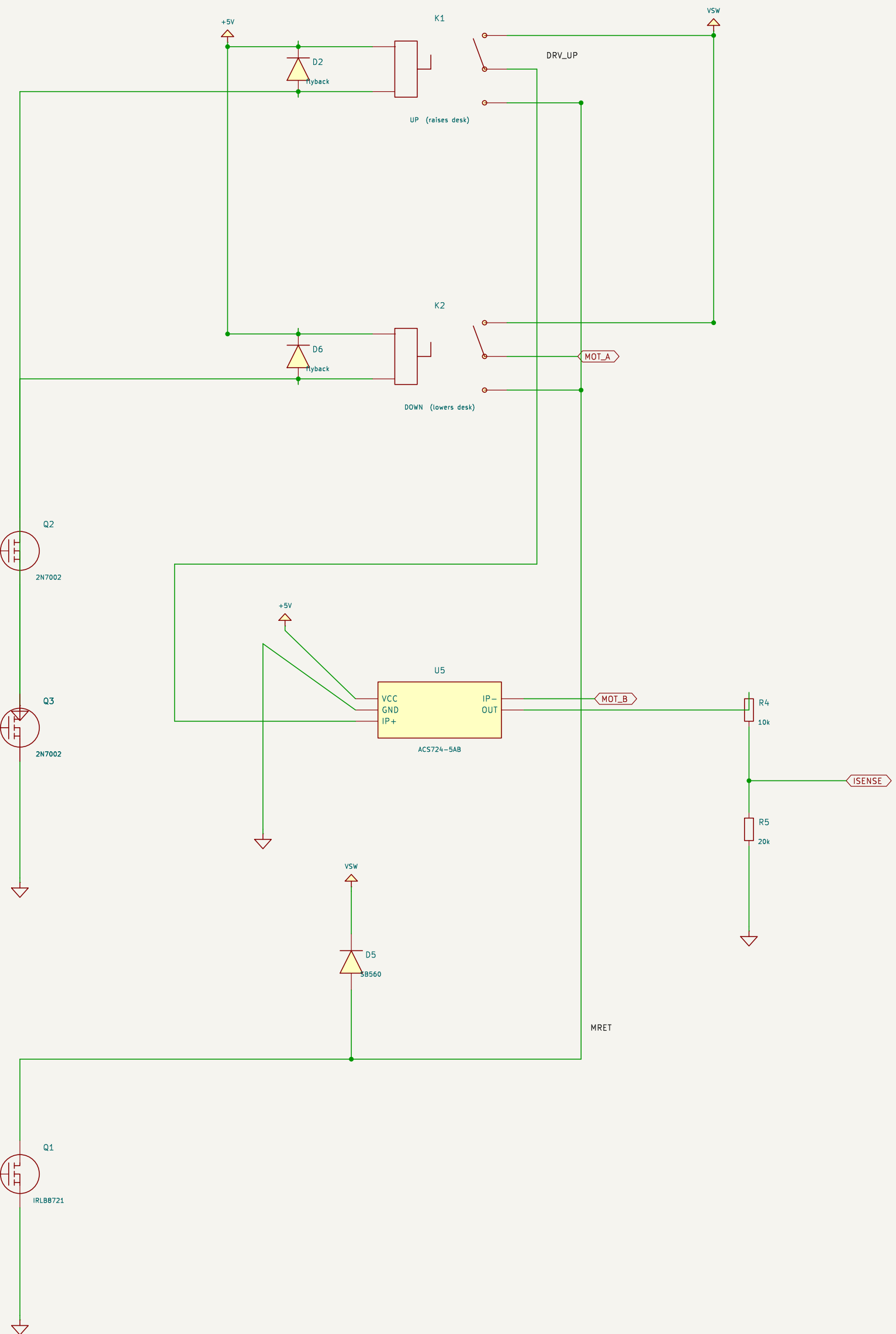
hold drive lines safe before boot

WATCHDOG + GATING

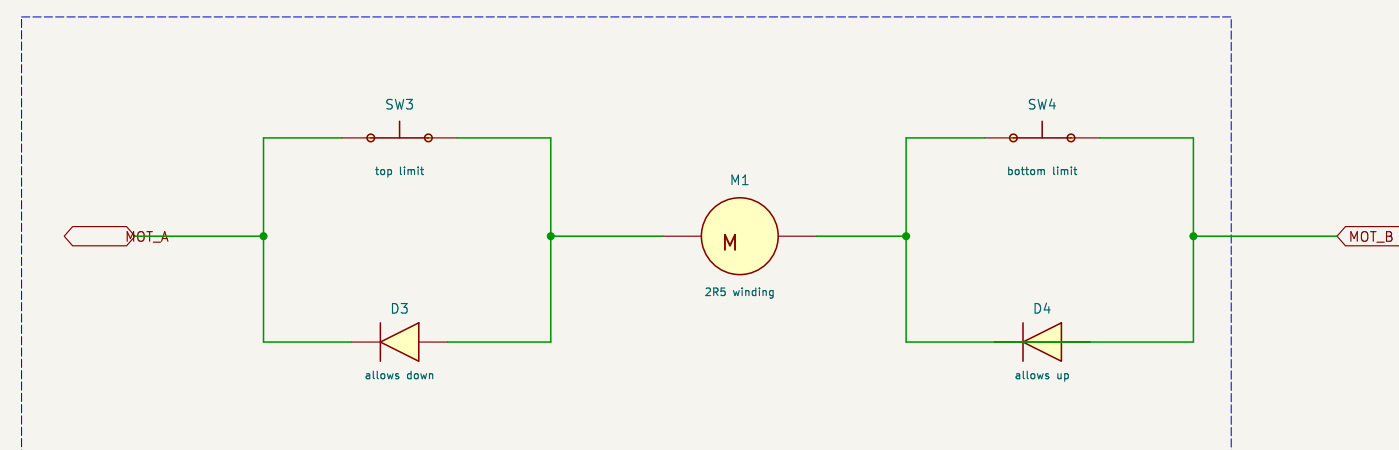


three gates of one 74HC08

DRIVE



LEG (unmodified)



SIMPLE IN-LINE DRIVE. The box sits in the 4-core cable and owns the winding permanently. Reverting is physical: motor – box – handset becomes motor – handset.

The handset is no longer in the motor circuit. It is fed 3V3 and its rockers are read as dry contacts – high at rest through the OEM NC contact, low when pressed. Because 29V can never appear on those conductors, no clamp diodes are needed.

POLARITY (measured on the 4-pin Mini-Fit Jr, circuits numbered on the housing):

circuit 1 = motor circuit 3 = supply +

circuit 2 = motor circuit 4 = supply -

circuit 2 positive, circuit 1 negative = UP. Reverse = DOWN.

MOT_A is defined as the conductor that must be POSITIVE to raise the desk (circuit 2). MOT_B is circuit 1.

K1 / K2 reproduce the OEM rest behaviour:

both off $\rightarrow$ both winding ends on VSW = BRAKE

K1 on -> MOT_B to return, MOT_A stays positive = UP